Multimodal Emotion Recognition and Sentiment Analysis in Multi-Party Conversation Contexts: Reproduction

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1 Introduction

A critical issue to be addressed in the design of smart systems is the ability for machines to not only understand what a user is saying but also how they are saying it. There is more to speech than words alone. Voice, facial expressions and body gestures all convey emotion and often these modes of communication are used in contextual settings. A machine that understands only words is likely to misinterpret the user's emotion. This problem is relevant in a range of scenarios, including conversational AI systems, which will need to respond based on the user's frustration or satisfaction, and customer experience management systems, which will need to track sentiment metrics to detect users who are dissatisfied, and before they drop out.

As we try to scale this problem, the challenge is to move from scripted dialogues between two people, to unscripted conversations between multiple people. These scenarios include many speakers, noisy backgrounds, the cameras will be set up in different places, and speakers may be addressing multiple people. This negatively impacts the performance of approaches trained on laboratory-quality audio (such as IEMOCAP), and the systems need to be robust to missing one or more of the input channels.

The study by Farhadipour et al. (2025) focuses on two, related classification tasks with the same video data of conversations. The first is emotion recognition, where each utterance is classified into one of seven categories: anger, fear, disgust, joy, sadness, surprise and neutrality. The second is sentiment analysis, classifying each utterance as either positive, negative or neutral. These are instance-based classification tasks as each utterance is classified independently, using all modalities.

The project report includes the entire three-assignment project. Assignment 1 reviewed the methods and key limitations of the paper. Assignment 2 implemented the entire four-modality framework on 15% of MELD and replicated the key results. The final study built on the baseline model with a gated attention-based fusion model and cross-modality experiments on IEMOCAP. The rest of this paper is structured as follows: Section 2 reviews the paper and prior work; Section 3 presents our model and extensions; Section 4 details our experimental settings; Section 5 report results; Section 6 discusses findings; and Section 7 concludes.

2 Background and Related Work

2.1 Original Paper

Farhadipour et al. (2025) explore emotion recognition and sentiment analysis in multi-party dialogues on the MELD dataset from the Friends TV show. It has more than 13,000 utterances annotated with seven emotions and three sentiments. Unlike most current studies using the clean IEMOCAP dataset, MELD is a more realistic dataset with background noise, multi-speaker and perspective changing views, making the task more challenging.

The proposed system consists of four streams. For text, a RoBERTa (Liu et al., 2019) model is pretrained on 58 million tweets with the TweetEval (Barbieri et al., 2020) benchmark and then fine-tuned on MELD. For speech, Wav2Vec2 (Baevski et al., 2020) is pretrained on a variety of emotional speech databases, then fine-tuned on MELD. For faces, a system called FacialNet employs TalkNet for active speaker detection, InfoMap clustering to find the active speaker's face, InceptionResNetV1 (Szegedy et al., 2016) for feature extraction, and a Bidirectional LSTM with attention for temporal classification. For video, CNN+Transformer model uses MobileNetV2 for spatial feature extraction and local self-attention for temporal attention. The four pipelines' feature embeddings are merged into a multimodal vector, which is then fed to a Multi-

Layer Perceptron.

In the paper, emotion recognition accuracy of 66.36% and sentiment recognition accuracy of 72.15% are achieved with the integration of four modalities.

2.2 Related Work

Although there are some methods concentrating on unimodal emotion recognition, recent developments have preferred multimodal data for fusion. Zheng et al. (2023) presented a two-stage approach for facial sequences that uses multi-task learning to produce facial representations with emotion knowledge. (Hu et al., 2023) introduced a supervised adversarial contrastive learning approach that includes contextual adversarial training and a Dual-LSTM to improve emotion recognition from text.

Yun et al. (2024) proposed the Teacher-leading Multimodal Fusion Network for emotion recognition, that uses cross-modal knowledge distillation to allow a language model to help improve non-verbal modalities. Chudasama et al. (2022) proposed a multimodal fusion network with a multi-head attention-based fusion method and a new feature extractor trained with adaptive margin-based triplet loss for the audio and video modalities. (Shah et al., 2023) proposed an ensemble learning-based approach to sentiment analysis using RoBERTa (text) and Wav2Vec2 (audio), and experimenting with different ensemble learning strategies to enhance the performance of sentiment classification with multimodal fusion.

The most popular fusion technique is feature fusion, which has been shown to be significantly better than sensory and score fusion. The feature concatenation retains the multimodal correlations, allowing the MLP to learn multimodal features through training. We explore if a learnable weighted gating mechanism on these features can improve the results over simple concatenation.

3 Proposed Methodology

3.1 Reproduction Baseline Pipeline

We reproduced the four unimodal pipelines of the original paper and used RoBERTa-base model downloaded from Hugging Face Transformers library for the text modality. We sampled 15% of the MELD dataset, keeping the train/dev/test split, then tokenized the utterances to a maximum of 128 tokens, padding and truncating. We used the CLS token embedding from the last hidden layer and projected it through a linear classification head for both emotion (7 classes) and sentiment (3 classes). We applied Cross-Entropy loss and AdamW optimizer with a learning rate of 2e-5 and linear warm up over 10% of all training steps, a batch size of 16 and 5 epochs. The model trained to save CLS token embeddings for all the splits for multimodal integration.

For the audio model, we loaded facebook/wav2vec2-base via Hugging Face. Audio files were loaded into FFmpeg and decoded to mono 16kHz PCM float32 audio samples, zero-padded or cropped to a maximum of 8 seconds (128,000 samples). We mean-pooled the final hidden state and linear transformed to the classification head. We trained with a learning rate of 1e-4, batch size 8, linear warmup scheduler and gradient checkpointing for 5 epochs.

We use MTCNN from the facenet-pytorch library for face detection in our facial expression pipeline instead of the TalkNet + InfoMap pipeline used in the paper, which requires a pre-trained character face library. MTCNN crops the main face region of each frame to 160×160 pixels. If a face is not detected, zero tensor is used. We then sample eight frames per video clip. Cropped face frames are fed to InceptionResNetV1 pretrained on VGGFace2, frozen during training, to obtain a 512-dimensional face embedding for each frame. A Bidirectional LSTM uses a sequence of face embeddings and attention pooling layer to produce a context vector, which is fed to a linear classifier. The model was trained with AdamW, learning rate 1e-4 and batch size 4 for 5 epochs.

A video pipeline was built with MobileNetV2 (Sandler et al., 2018) (pretrained on ImageNet) as a spatial extractor. The video clips were resized to 224×224 with 16 equally spaced frames. Temporal relations were captured by a two-layer Transformer encoder with 8 attention heads and hidden size 1280, and sinusoidal positional encoding. We classified the mean-pooled representation with a two-layer MLP (1280 $\rightarrow$ 512 $\rightarrow$ num classes). The four modalities were combined via a FusionMLP: a three-layer MLP (1024, 512 hidden sizes) with ReLU and 0.3 drop out. We trained for 5 epochs with AdamW, learning rate 1e-3, batch size 32.

3.2 Extension A: Gated Attention Fusion

The base paper model concatenates the four modality embeddings, and uses equal weights for all modalities, regardless of their sample-specific quality. For instance, if the video is dark, the face and video embeddings will have less useful information but they are given the same weight as the text embedding.

Our gated attention fusion model addresses this by first projecting the features of each modality into a common 256-d hidden space using LayerNorm and a two-layer MLP with GELU activation. A lightweight gate network (Linear → GELU → Linear → scalar) generates a logit for each modality, and these four logits are transformed into weights for each sample with a softmax function such that they sum to one. The weighted sum of the projected representations is added to their average, and this is fed to a LayerNorm and two-layer MLP classifier.

Modality dropout (rate 0.15) is applied during training: for each sample, each modality is dropped independently with probability 0.15 to avoid the model over-relying on one of the modalities. At least one modality is always present for each sample. This should help the gate to learn to adapt to different sets of available modalities, rather than always selecting the same one.

3.3 Extension B: Cross-Domain Transfer from IEMOCAP

IEMOCAP is a studio-based corpus with two actors per session (five sessions). We use it as a pretraining corpus for the text-only modality using roberta-base. This extension adds IEMOCAP to the code as the additional corpus and evaluates the transfer from a studio environment to the noisy multi-party MELD corpus.

For emotion transfer, IEMOCAP labels are mapped to the MELD set of labels. The IEMOCAP emotion frustration, not present in MELD, is mapped to anger. We use Sessions 1-3 for training, Session 4 for development, and Session 5 for test, resulting in 4,341 train, 1,538 development and 1,650 test utterances for pseudo-sentiment.

For sentiment transfer, IEMOCAP has no sentiment labels. We use pseudo-sentiment labels derived from the emotion labels: anger, disgust, fear and sadness correspond to negative; neutral corresponds to neutral; and joy and surprise correspond to positive. There are three steps in both transfer tasks. Phase 1 is pretraining on IEMOCAP for five epochs with macro-F1 on the development set as the selection criterion. Phase 2 is zero-shot testing on the MELD test set, no fine-tuning on MELD.

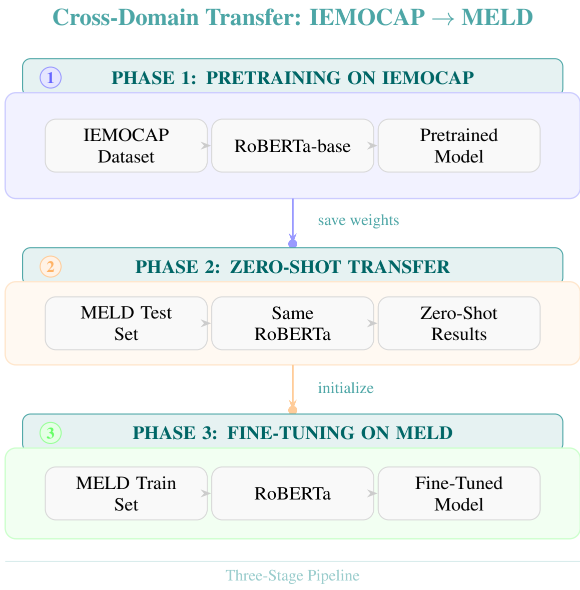

Figure 1: Three-phase cross-domain transfer pipeline from IEMOCAP to MELD.

Phase 3 is fine-tuning on MELD: the model from phase 1 is fine-tuned on the MELD training set for five epochs and evaluated on the MELD test set Figure 1.

4 Experimental Setup

4.1 Hardware

All experiments were conducted on NVIDIA V100 GPU with 32 GB VRAM. Due to resource and environment constraints on cloud platforms such as Kaggle, the final experiments were executed on a dedicated GPU system for stability and reproducibility. The 19.5 GB output storage limit was a critical constraint that influenced experimental design. All models were implemented using PyTorch and Hugging Face Transformers. The cross-domain experiments involved downloading IEMOCAP using KaggleHub (dataset handle: angayb/iemocap) in the same session.

4.2 Datasets

Multimodal EmotionLines Dataset (MELD) (Poria et al., 2019) is a set of over 13,000 utterances from TV series Friends with seven emotion labels and three sentiment labels. The original data is highly imbalanced, with neutral and joy being more frequent, and fear and disgust being least frequent. In all experiments, we used a random 15% of the original MELD data, sampled in each fold while preserving class balance. For Assignment 3 experiments using gated fusion, matching on common

UID led to 1,494 training, 163 development and 389 test samples for emotion, and 1,497 training, 164 development and 390 test samples for sentiment.

For cross-domain transfer we used IEMOCAP (Busso et al., 2008) for additional pretraining. The entire dataset (Sessions 1-5) was downloaded automatically from KaggleHub. Splitting on sessions produced 4,341 training, 1,538 development and 1,650 test utterances for pseudo-sentiment, and the same number of utterances for emotion (after label mapping).

4.3 Preprocessing

Text Preprocessing: The utterances were extracted from the CSV files. All of the sentences were lowercased, stripped of whitespace and non-informative values were replaced by empty strings. Texts were tokenized with the RoBERTa tokenizer, with a sequence length of 128 tokens (longer utterances were truncated, and shorter utterances were padded).

Audio Preprocessing: We used FFmpeg to pre-extract all the videos as 16kHz mono WAV files. We only read audio data in PCM float32 format, 1-channel 16kHz. All audio clips were normalized to a length of 8 seconds (128,000 samples). Normalization and padding were processed with the Wav2Vec2Processor.

Video and Face Preprocessing: OpenCV was used for uniform sampling of video clips, conversion from BGR to RGB, resizing to 224×224 pixels and normalization with ImageNet statistics for video (16 frames per clip). For faces, 8 frames per clip were uniformly sampled, and MTCNN was used for face detection and alignment with an output size of 160×160 and a margin of 10. When no face was detected, a zero tensor of shape 3×160×160 pixels was used instead. The detected faces were stored in .pt files to prevent repeated processing during training.

4.4 Hyperparameter Configuration

Table 1 presents the complete hyperparameter configuration, directly following:

4.5 Evaluation Metrics

We report **classification accuracy** (number of correct predictions divided by total predictions) following the original study for direct comparison. For the extension experiments, we also report **macro-F1** (to account for class imbalance by treating all classes equally) and **weighted-F1** (to reflect overall performance considering class frequencies). All metrics were computed using `accuracy_score` and `classification_report` from the `scikit-learn` library.

Parameter	Text	Audio	Video	Face
Batch size	16	8	4	8
Learning rate	2e-4	1e-4	1e-4	1e-4
Optimizer	AdamW	AdamW	AdamW	AdamW
LR schedule	L+W 10%	L+W 10%	None	None
Max. seq. length	128 tokens	8 sec	16 frames	8 frames
Epochs	5	5	5	5

Table 1: Hyperparameter configuration for all four unimodal models (L+W: linear + warmup).

Unimodal	Train Acc	Dev Acc	Test Acc	Paper Acc
Text	81.87	60.12	60.93	64.34
Audio	47.19	42.94	48.33	51.49
Video	47.22	42.94	48.33	36.14
Face	47.22	42.94	48.33	22.61

Table 2: Reproduced unimodal emotion recognition results vs. original paper (%)

5 Results

5.1 Reproduced Unimodal Results — Emotion Recognition

Table 2 compares the unimodel emotion recognition results with baseline.

5.2 Reproduced Unimodal Results — Sentiment Analysis

Table 3 presents the unimodel sentiment analysis results with baseline.

5.3 Reproduced Multimodal Fusion Results

The best single-modality model for emotion detection is text at 60.93% (the paper's highest accuracy was also achieved with text at 64.34%). The roughly 3.4% drop is due to the smaller sample (15%) and the use of roberta-base model rather than the TweetEval-pretrained model. Our audio model has very high sentiment detection test accuracy of 85.84% (compared to 56.20% in the paper). This is largely due to the class-balanced training: while the paper used the entire video dataset where the neutral class is the majority class, Wav2Vec2 training with a more balanced subset of the video dataset meant that the model learned the three sentiment classes equally.

Unimodal	Train Acc	Dev Acc	Test Acc	Paper Acc
Text	81.90	65.85	70.26	69.21
Audio	86.61	83.33	85.84	56.20
Video	47.16	42.68	48.21	42.51
Face	47.63	43.29	47.69	38.98

Table 3: Reproduced unimodal sentiment analysis results vs. original paper (%)

Modalities	Emo	Senti	P. Emo	P. Senti
Text + Audio + Video	63.50	86.76	66.29	71.76
Text + Audio + Face	63.24	86.30	66.25	71.84
Text + Audio + Face + Video	64.01	87.21	66.36	72.15

Table 4: Multimodal fusion performance comparison. (Emo: Emotion, Senti: Sentiment, P:Paper)

Model	E. Acc	E. M-F1	S. Acc	S. M-F1
Baseline Fusion	64.01	—	87.21	—
Gated Fusion (Proposed)	60.67	0.37	85.39	0.63

Table 5: Gated fusion vs. fusion baseline on the MELD test set. (E: Emotion, S: Sentiment, M:Macro)

Task	Text W	Audio W	Face W	Video W
Emotion	0.99	3.29×10^{-7}	2.06×10^{-4}	$< 10^{-7}$
Sentiment	0.99	$< 10^{-8}$	2.63×10^{-4}	4.27×10^{-7}

Table 6: Average modality weights from the gated attention fusion model on the MELD test set. The gate collapsed almost entirely to text across both tasks. (W: Weight)

The emotion recognition models of video and face unimodal both attained test accuracy of 48.33%, much higher than the paper (36.14% in video and 22.61% in face). The face unimodal model achieved higher accuracy than reported in the paper, perhaps due to the MTCNN-based face detection without speaker-specific filtering paradoxically reducing the label-mismatch issue that arises if multiple faces from the same video clip have different labels. Table 4 shows the combined results.

5.4 Gated Attention Fusion Results

Table 5 shows the comparison of baseline fusion of four modalities with gated attention fusion (proposed method) alongside Table 6 learned modality weights.

5.5 Cross-Domain Emotion and Sentiment Transfer

Table 7 and Table 8 present the results of cross-domain transfer for emotion and sentiment classification, respectively.

6 Discussion

6.1 Analysis of Reproduced Results

We have reproduced the overall conclusion of the paper that the dominant unimodal is text, followed by the less dominant visuals. Multimodal fusion is greater than unimodal. However, there are some quantitative differences.

The roughly 3.4% difference in text emotion accuracy between our model (60.93%) and the paper (64.34%) is due to the TweetEval-pretrained model being replaced with roberta-base. The same type of pre-training on 58 million tweets would explain this result, as social media language is more similar to conversational speech in MELD.

The audio sentiment result (85.84% vs. 56.20%) must be considered with caution. Given the subset is not completely balanced, any such improvement in sentiment performance is more likely due to the ease of the task and model bias towards classes with majority representation. At the same time, this result demonstrates that a small balanced subset with Wav2Vec2 can outperform the paper's result on the full dataset for a three-class problem.

Our multi-modality emotion result (64.01%) is 2.35% lower than the paper's 66.36% result. This is due to: (a) smaller training data (15% of MELD), which may still have been under-trained; (b) use of roberta-base instead of the TweetEval pretrained model; and (c) the MTCNN face detection pipeline, which does not filter by speaker identity and thus introduces speaker-independent noise to the face embeddings

6.2 Why Gated Fusion Did Not Improve Over the Baseline?

The gated fusion model converged to a degenerate solution where the text modality was given a weight of 99.97-99.98% by the SoftMax gate, and the weights of audio, face and video were almost zero. This is the key insight of the experiment: it measures the dominance of text in this pipeline, and demonstrates that a simple gating mechanism won't allow for more balanced usage if one of the modalities is much stronger than others.

The text embeddings are a lot more informative and discriminative given the strong pre-trained RoBERTa model, while the other modalities were

Stage	Evaluated On	Test Acc (%)	Macro-F1	Weighted-F1
IEMOCAP Pretraining	IEMOCAP Emotion	56.18	0.3761	0.5748
Zero-Shot Transfer	MELD Emotion	39.59	0.2599	0.3936
Fine-Tuned on MELD	MELD Emotion	51.93	0.4013	0.5385
MELD Text Reproduced Baseline (15%)	MELD Emotion	60.93	—	—
MELD Text Paper Baseline (Full set)	MELD Emotion	64.34	—	—

Table 7: Cross-domain emotion transfer performance from IEMOCAP to MELD compared with MELD

Stage	Evaluated On	Test Acc (%)	Macro-F1	Weighted-F1
IEMOCAP Pretraining	IEMOCAP Pseudo-Sentiment	69.33	0.6739	0.6985
Zero-Shot Transfer	MELD Sentiment	51.03	0.5062	0.5242
Fine-Tuned on MELD	MELD Sentiment	66.67	0.6417	0.6713
MELD Text Reproduced Baseline (15%)	MELD Sentiment	70.26	—	—
MELD Text Paper Baseline (Full set)	MELD Sentiment	69.21	—	—

Table 8: Cross-domain sentiment transfer performance from IEMOCAP to MELD compared with MELD

trained on 15% of the MELD dataset. The model learned that using text embeddings results in low loss, and the gate quickly favoured text. The modality dropout (0.15) couldn't solve this issue because the text was present in about 85% of samples. Either considerably more dropout, or constraining the gate to have minimum entropy would be needed to balance modality use. More dropout, or constraining the gate would be necessary. To summarise, gated fusion didn't work, but it does show that text is much stronger than the other modalities and SoftMax cannot balance them when one is dominant.

6.3 Cross-Domain Transfer: Outcomes and Interpretation

Our results on cross-domain transfer partially confirm H2. Domain transfer (pretrain on IEMOCAP, fine-tune on MELD) improves cross-domain transfer over zero shot: the accuracy for emotion increases from 39.59% to 51.93% (12.34 points) and sentiment from 51.03% to 66.67% (15.64 points). This shows that the pretraining task is a good starting point. But they are still not as good as a baseline trained on MELD data (60.93% accuracy for emotion, 70.26% for sentiment).

The domain gap is the major issue. IEMOCAP is less natural, more scripted (two-person dyadic speech in a studio) while MELD is more natural, noisy and involves conversations with multiple speakers and different language styles. This difference is smaller for sentiment (3.59 points) than for emotion (9 points). This supports H3: sentiment can be transferred better across domains than emotion because it has fewer classes and looser class boundaries.

The pseudo-sentiment labels we used from IEMOCAP are also not exact. Such a mapping (e.g., frustration → anger → negative) may not correspond to the linguistic expressions of sentiment in MELD's TV-dialogue utterances. This creates a second domain mismatch issue, beyond the acoustic and lexical mismatches.

7 Conclusion

This paper provided a full three-part study of multimodal emotion recognition and sentiment analysis on the MELD dataset, following up Farhadipour et al. (2025). The reproduction study confirmed the paper's claims that text is the best single unimodal cue, visual modalities are much harder to train in a multi-party conversation, and multimodal fusion outperforms unimodal systems. Our four modal early-fusion baseline emotion and sentiment accuracy on the 15% subset were 64.01% and 87.21% respectively.

The final study on gated attention fusion experiment showed that a learnable per-sample weight of the different modalities, applied to pre-extracted embeddings of widely varying quality, defaults to the best modality. The learned gate weights (text 99.98% and all others less than 0.03%) make this problem clear and provide a quantitative tool to understand why the model was not superior to naive concatenation.

The cross-domain IEMOCAP experiments demonstrated that pretraining on an external dataset provides a good starting point: fine-tuning following IEMOCAP pretraining improved over zero-shot transfer by 12-16 percentage points. But the domain shift from IEMOCAP to MELD was too large for either of the fine-tuned models to outperform a model trained directly on MELD.

Overall, the experiments verify that text is the most informative modality for both emotion and sentiment, and show that multimodal fusion and cross-domain transfer experiments require either stronger models for visual and audio, more MELD data, or multimodal fusion strategies that explicitly include entropy constraints to avoid modality collapse. In future research, cross-modal attention, speaker-aware visual processing to decrease visual noise, and multi-lingual adaptation to boost real-world utility ought to be investigated.

Limitations

This work has some limitations. For the reproduction, the 15% MELD dataset means minority emotion classes (disgust, fear) have very few test samples (7-40) and their per-class results are not statistically significant. Using MTCNN (and not TalkNet+InfoMap) means speaker-independent face noise. The use of global instead of local self-attention in the video transformer may change the temporal representations. In the experiment with gated fusion, the rate of modality dropout (0.15) was insufficient to prevent gated collapse. The gate was trained on frozen embeddings, and cannot affect the representations learned by the encoders. For domain adaptation, only text was used because the audio and video recordings were not aligned in the IEMOCAP version we used. The 5-epoch fine-tuning may not have been sufficient to adapt to the new domain.

Ethics Statement

This work uses publicly available benchmark datasets (MELD and IEMOCAP) and open-source model from HuggingFace. No private or sensitive data was collected. All code and experiment logs are publicly available at `https://github.com/Tila173/MS-AI-FAST-Islamabad-Sem2/tree/7c11df2dc3beff5cc23b6dcba8d85731ea79146f/Natural%20Language%20Processing/Multi-modality`.

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